

# MD ABU SAYEB RAYHAN

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## Research Interests

Robot perception, embodied AI and intelligent autonomous systems; computer vision (multi-object tracking, activity recognition, pose estimation), multimodal vision-language models and generative models for simulation and robotic learning.

## Education

**Bachelor of Science**, Computer Science & Engineering GPA: 3.4 / 4.0 Jan 2019 – May 2024  
Comilla University, Bangladesh

**Relevant Coursework:** Artificial Intelligence, Algorithm Design, Theory of Computation, Introduction to Data Science, Object-Oriented Programming, Research-Based Project.

**Bachelor's Thesis:** *Text-to-Image Generation using Diffusion Models*. Fine-tuned Stable Diffusion (U-Net backbone) on a custom dataset of Bangladeshi landmarks. Investigated prompt engineering and conditioning mechanisms for generative fidelity.

## Publications

- **Forensic Invariant Learning for Synthetic Image Detection: Bridging Benford's Law and Topological Analysis with Machine Learning Ensembles.** [IEEE Xplore](#)  
Md Abu Sayeb Rayhan, Ajoy Shutrathar and Hustagna Hiya
- **RGEF: Reliability-Gated Evidential Fusion for Robust Medical AI.** (Accepted to IEEE RAAICON 2026)  
Md Abu Sayeb Rayhan, Mosammat Fahimah, Md Tarek Mahmud, Hustagna Hiya and Ajoy Shutra Dhar
- **BD Eye-Doc: When Computer Vision Meets Conversational AI for Scalable Eye Disease Screening in BD.** [IEEE Xplore](#)  
Ajoy Shutrathar, Md Abu Sayeb Rayhan, Mahmuda Khatun and Bishwajit Das
- "Is CNN Dead? A Comprehensive Empirical Analysis of Convolutional Neural Networks, Ensemble Methods and Vision Transformers for Rice Leaf Disease Classification in Low-Resource Agricultural Settings." *Presented – ECCT 2026*. To appear in ACSR, Atlantis Press (Springer Nature).

## Research Experience

**Research Assistant** – Comilla University Supervisor: Partha Chakraborty, Associate Professor Jan 2023 – Mar 2024

- Contributed to empirical studies in **Human-Robot Interaction (HRI)**: prepared datasets, conducted experiments and performed quantitative analysis.
- Prepared and validated results subsequently incorporated into peer-reviewed conference publications.

## Professional Experience

**Data Scientist** – Othello Financial Service Co. Ltd., Japan Oct 2024 – Mar 2026

- Investigated **multi-object tracking, pose estimation and human activity recognition** for automated retail behaviour analysis.
- Designed a spatial clustering pipeline using **MoveNet** and **TakumiVision** for group-dynamics modeling, reducing processing latency by approximately 70% through parallel inference optimization.
- Developed a video event-segmentation system using a fine-tuned **YOLO-based detector** (90%+ accuracy); automated highlight extraction cut NFT production cost by 50%.
- Implemented a **RAG-based conversational system** (LangChain, Pinecone, GPT-4) for context-aware document querying; deployed on Microsoft Azure.
- Worked on **edge AI** for automated retail data acquisition feeding real-time computer vision pipelines.

**AI Engineer** – Bangladesh Government Project (Contractual) Feb 2024 – Jul 2024

- Developed a multilingual (Bangla/English) **NLP + OCR** pipeline for keyword and context-based document retrieval across PDF, Word, Excel and image formats; built as a desktop application (PyQt + ChromaDB).
- Coordinated implementation within a three-person team across full product lifecycle from design to deployment.

## Technical Skills

- **Programming:** Python (advanced), C/C++, SQL, Java
- **ML / AI Frameworks:** PyTorch, TensorFlow, OpenCV, MMAAction2, YOLO, ByteTrack, MoveNet, Stable Diffusion, LangChain, Hugging Face, Scikit-learn, Pandas
- **Methods:** Object Detection, Action Recognition, Pose Estimation, Spatial Clustering, Diffusion Models, OCR, RAG, Dense Embeddings, Temporal Segmentation, NLP
- **Infrastructure:** Microsoft Azure, Docker, CI/CD, Git; Flask, PyQt; Pinecone, ChromaDB, MongoDB, Azure SQL; Power BI

## Honours & Certifications

- **Best Presenter Award**, IEEE QPAIN 2026
- Participant, Inter-University Programming Contest, Comilla University (2019–2021)
- Certified, *Skill Development for Mobile Game & Application Development* – ICT Division, Government of Bangladesh (2022)
- Served as a Joint Secretary, North Bengal Student Association, Comilla University(2023-2024)

## Language Proficiency

English – Fluent | Bangla – Native | Japanese – Basic Daily Conversation (JLPT N4 equivalent)

## References

**Partha Chakraborty**  
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